Analyzing and Improving CLIP for Image–Text Retrieval

Wei Luo Haosen Fang

1 Project Description and Goals

This project aims to analyze the behavior of the CLIP model on image–text retrieval tasks and develop simple yet effective methods to improve its performance. CLIP is a widely used vision-language model that aligns images and text in a shared embedding space, enabling zero-shot classification and retrieval. While CLIP performs well on many benchmarks, it can fail in cases involving ambiguity, complex scenes, or unusual visual contexts.

The primary objectives of this project are:

1. Evaluate CLIP's baseline performance on a standard dataset.
2. Identify and categorize its failure modes.
3. Propose and implement lightweight improvements to enhance retrieval performance.

1.1 Minimum Goals

- Implement image-to-text and text-to-image retrieval using CLIP.
- Evaluate performance using standard metrics such as Recall@K.
- Identify and document at least three categories of failure cases.

1.2 Maximum Goals

- Improve retrieval performance through techniques such as prompt engineering and reranking.
- Compare multiple improvement strategies and analyze their effectiveness.
- Provide a systematic analysis of how different prompts affect model behavior.

2 Member Roles

- **Wei Luo**
  - Implement baseline CLIP retrieval pipeline
  - Conduct experiments and evaluation
  - Develop improvement methods (prompt engineering, reranking)
- **Haosen Fang**

  - Dataset preprocessing and management
  - Visualization and result analysis
  - Assist with report writing and figure generation

Both members will collaborate on experiment design, analysis of results, and writing the final report.

3 Resources

3.1 Datasets

We plan to use one of the following publicly available datasets:

- **Flickr30k**: Contains images with multiple human-annotated captions.
- **MS COCO**: A large-scale dataset widely used for image-caption tasks.

3.2 Implementation Platform

- Python with PyTorch
- Pretrained CLIP model (ViT-B/32 variant)
- GPU access (local machine or cloud if necessary)

3.3 External Code

We will use the official CLIP repository and adapt it for retrieval tasks. Any external code will be clearly documented and cited.

4 Methodology

4.1 Baseline

We will implement a retrieval system using CLIP embeddings:

- Encode images and text into a shared embedding space.
- Compute cosine similarity to rank results.
- Evaluate using Recall@K metrics.

4.2 Failure Analysis

We will analyze cases where CLIP performs poorly, such as:

- Ambiguous or polysemous queries (e.g., "apple")
- Images with multiple objects
- Abstract or unusual visual scenes

4.3 Proposed Improvements

We will explore the following methods:

- **Prompt Engineering**: Modify input text prompts (e.g., "a photo of a _____") to improve alignment.
- **Reranking**: Refine top-K retrieval results using additional scoring strategies.

5 Reservations (Potential Challenges)

- **Dataset Complexity**: Large datasets like MS COCO may require significant preprocessing and computation time.
- **Limited Improvement Margin**: CLIP is already strong; improvements may be incremental.
- **Evaluation Design**: Ensuring fair and consistent comparison between methods may be challenging.

To mitigate these risks, we will:

- Start with a smaller dataset (e.g., Flickr30k) if needed.
- Define a clear minimum goal that ensures a complete project even if improvements are modest.

6 Relationship to Background

The project aligns well with our background in computer vision and machine learning. We have prior experience with:

- Image processing and feature extraction
- Deep learning frameworks such as PyTorch
- Basic computer vision tasks (e.g., denoising, segmentation)

This project extends our knowledge into vision-language models and multimodal learning. It also provides an opportunity to explore practical techniques for improving model performance without requiring large-scale training.

7 Expected Outcome

By the end of this project, we expect to:

- Gain a deeper understanding of CLIP's strengths and limitations.
- Develop practical methods to improve image–text retrieval.
- Produce a comprehensive report with quantitative results and visual examples.

Even if improvements are modest, a thorough analysis of failure cases and experimental results will demonstrate meaningful insight into the model's behavior.